Code No: 136FC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
FUNDAMENTALS OF ROBOTICS
(Common to CE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	What is precession of movement?	[2]
b)	Give one application of proximity sensor.	[3]
c)	What is pivoting movement in the gripper mechanism?	[2]
d)	What are adhesive grippers?	[3]
e)	What is homogeneous transformation matrix?	[2]
f)	Differentiate forward and inverse kinematics.	[3]
g)	What are the inputs for inverse kinematics problem?	[2]
h)	Explain joint movement method to define position in space.	[3]
i)	What are front and back lighting in machine vision?	[2]
j)	What is textual robot language in robot programming methods?	[3]

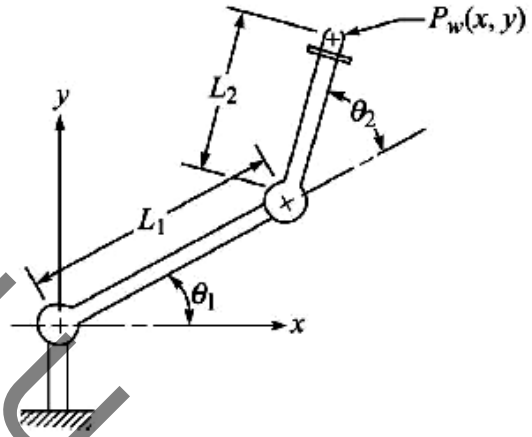
PART – B

(50 Marks)

2.a)	What are the various components of the industrial robotics?	
b)	What are the various types of joints used in robotic systems?	[5+5]
	OR	
3.a)	Differentiate parallel and serial manipulators.	
b)	Briefly describe about the torque sensor and its applications.	[5+5]
4.a)	Analyse the gripper force used in the gripper mechanism at the wrist end.	
b)	Discuss the gear and rack method of actuating the gripper.	[5+5]
	OR	
5.a)	What are the advantages of magnetic gripper?	
b)	How the vacuum cups can be used for handling the objects?	[5+5]

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6. Give the forward and reverse transformation of a 2 DoF arm of type RR, given its position. [10]



OR

7. A vector v=i+5j+2k is rotated by 60^0 about the Z axes, followed by 30^0 about X axes and then translated by 2,4,6 units in the x, y and z directions respectively. Find the final vector with reference to the reference frame. [10]

8. Discuss and calculate the various parameters concerned with 3-5-3 trajectory method. [10]

OR

9. Describe the features of servo system and control architecture for robot systems. [10]

10.a) How does the teach pendant performs the operations.
b) Explain the object recognition technique in the image processing and analysis. [5+5]

OR

11.a) Explain the feature extraction technique in the image processing and analysis.
b) What are the basic three applications of machine vision? [5+5]

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